Homework #5

5.1 Calculate the output resistance and the minimum output voltage, while maintaining all devices in saturation, for the circuits shown in Figure 5.1. Assume that i_{OUT} is actually 10μA. Use Table 5.1 for device model information. V_{bs}=0 V.

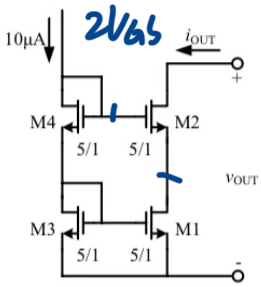


Fig 5.1

$2V_{GS}$

$V_1 = V_{GS3} + V_{GS4} - V_{GS2} = V_{GS}$

$V_T = V_{T0} = 0.7V$

$\lambda = 0.1V^{-1}$

$K = \mu C_{ox} = 134$

解: $i_{out} = 10\mu A \Rightarrow r_{O1} = r_{O2} = \frac{1}{\lambda i_{out}} = 1\times10^6\,\Omega$

$I_D \approx \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2$

$$\Rightarrow \begin{cases} V_{GS} = \sqrt{\frac{2I_D}{\mu_n C_{ox}\frac{W}{L}}} + V_{TH} = 0.873V \\ g_m = \frac{\partial I_D}{\partial V_{GS}} = \mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH}) = \sqrt{2I_D\mu_n C_{ox}\frac{W}{L}} = 1.16\times10^{-4}S \end{cases}$$

$r_{out} = r_{O1} + r_{O2} + g_m r_{O1} r_{O2} = 1.13\times10^3\,\Omega = 113M\Omega$

$V_{outmin} - V_{GS} = 2V_{GS} - V_{GS} - V_{TH}$

$\Rightarrow V_{outmin} = 2V_{GS} - V_{TH} = 1.046V$

5.2 A reference circuit is shown in figure 5.2, assume that $(W/L)_1=(W/L)_2=(W/L)_3=4$, $(W/L)_4=1$, please calculate the symbolic expression of V_{REF}.（已知各管处于饱和区且各管阈值电压为Vt）

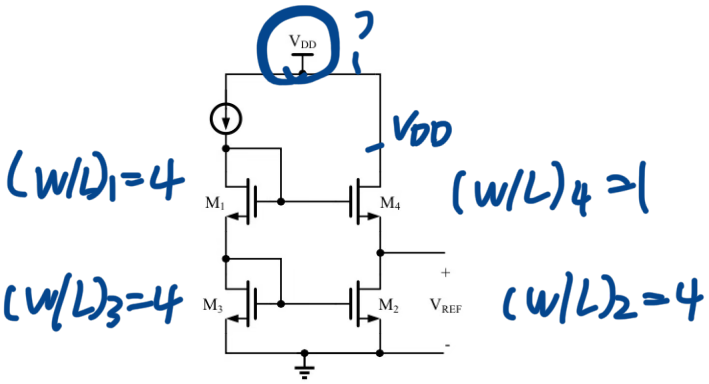


Figure 5.2

解 $V_{OD}=V_{GS}-V_{TH}=\sqrt{\frac{2I_D}{\mu_n C_{ox}\frac{W}{L}}}$

故 $V_{OD1}=V_{OD2}=V_{OD3}=\frac{1}{2}V_{OD4}$

故 $V_{REF}=V_{GS3}+V_{GS1}-V_{GS4}$

$=V_{OD3}+V_{TH}+V_{OD1}+V_{TH}-(V_{OD4}+V_{TH})$

$=V_t$

5.3 As the circuits shown in Figure 5.3, I_{REF}=0.3mA and γ=0. Using the model parameters in Table 5.1,

(a) Calculate the voltage V_b when V_X=V_Y.

(b) If V_b is 100mV smaller than the value in (a), calculate the deviation of I_{out} from 300 μ A.

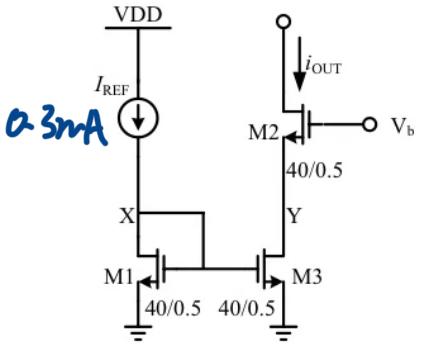


Figure 5.3

$\lambda = 0.1V^{-1}$

$K = 134\mu A/V^2$

解: (a) $V_X = V_Y$ 时, $I_{out} = I_{REF} = 0.3\,mA$

$$I_D \approx \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2$$

$$\Rightarrow V_{GS} = \sqrt{\frac{2I_D}{\mu_n C_{ox}\frac{W}{L}}} + V_{TH} = 0.937V$$

$$\Rightarrow V_b = 2V_{GS} = 1.873V$$

(b) $$I_{REF} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2[1+\lambda V_{GS}]$$

$$I_{out} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2[1+\lambda(V_{GS}-V_b)]$$

$\lambda(L=0.5M) = 2\lambda(L=1M) = 0.2V^{-1}$

故 $$I_{out} = I_{REF}\frac{1+\lambda(V_{GS}-V_b)}{1+\lambda V_{GS}}$$

$$\Delta i_{out} = I_{REF}\frac{-\lambda V_b}{1+\lambda V_{GS}} = -5.05\mu A$$

Iout + Swing

5.4 Design M3 and M4 of Figure 5.4(a) so that the output characteristics are identical to the circuit shown in Figure 5.4(b). It is desired that i_{OUT} is ideally 10uA.

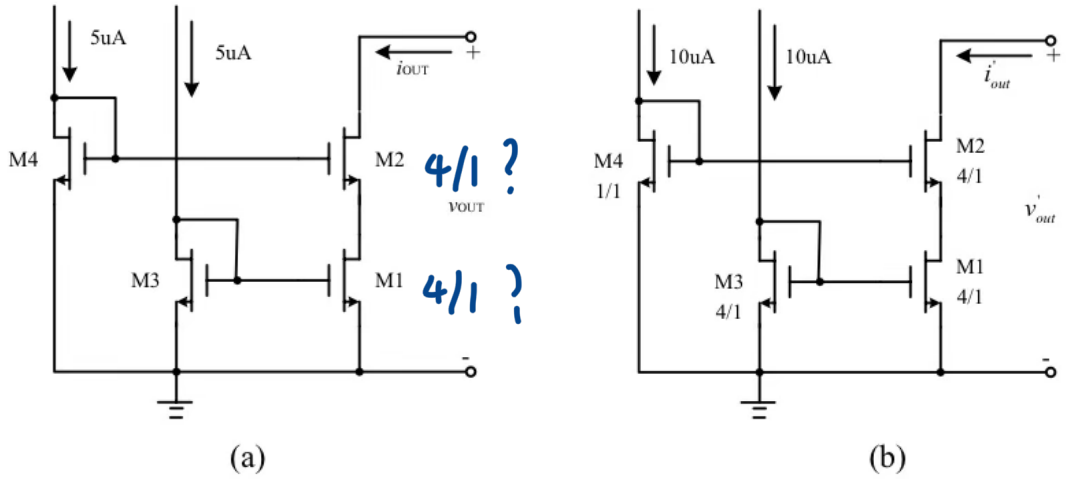


Figure 5.4

解. $I = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2 \propto \frac{W}{L}$

$\frac{I_{out}}{I_{REF}} = 2 \Rightarrow \frac{(W/L)_{out}}{(W/L)_{REF}} = 2$

故 $(\frac{W}{L})'_3 = \frac{2}{1}$

$V_{OD} = \sqrt{\frac{2I}{\mu_n C_{ox}\frac{W}{L}}} \propto \sqrt{\frac{I}{W/L}}$

$V_{out,min} - (V_{GS4}-V_{GS2}) = V_{GS2}-V_{TH}$

$\Rightarrow V_{out,min} = V_{GS4}-V_{TH} = V_{OD4}$

$V'_{out,min} = V_{out,min}$, 故 $V'_{OD4} = V_{OD4}$

又 $I'_{M4} = \frac{1}{2}I_{M4}$, 故 $(\frac{W}{L})'_4 = \frac{1}{2}(\frac{W}{L})_4 = \frac{1}{2}$